

ENGINEERING SPRINT 10

LWC Component Communication and Student Profile Application

1. Introduction:

Engineering Sprint 10 – Overview

Engineering Sprint 10 focuses on developing **Lightning Web Components (LWC)** and understanding how different components work together.

The main concepts covered were:

- Parent-to-Child Communication
- Child-to-Parent Communication
- Public Properties and Custom Events
- Job Card Component
- Student Profile Component
- Form Handling and Validation
- Apex Integration
- Saving and Loading Records
- Success and Error Handling
- Reusable Components

The main objective was to learn how **LWC components communicate with each other and interact with Salesforce data** to build a simple and functional application.

2. Component Architecture:

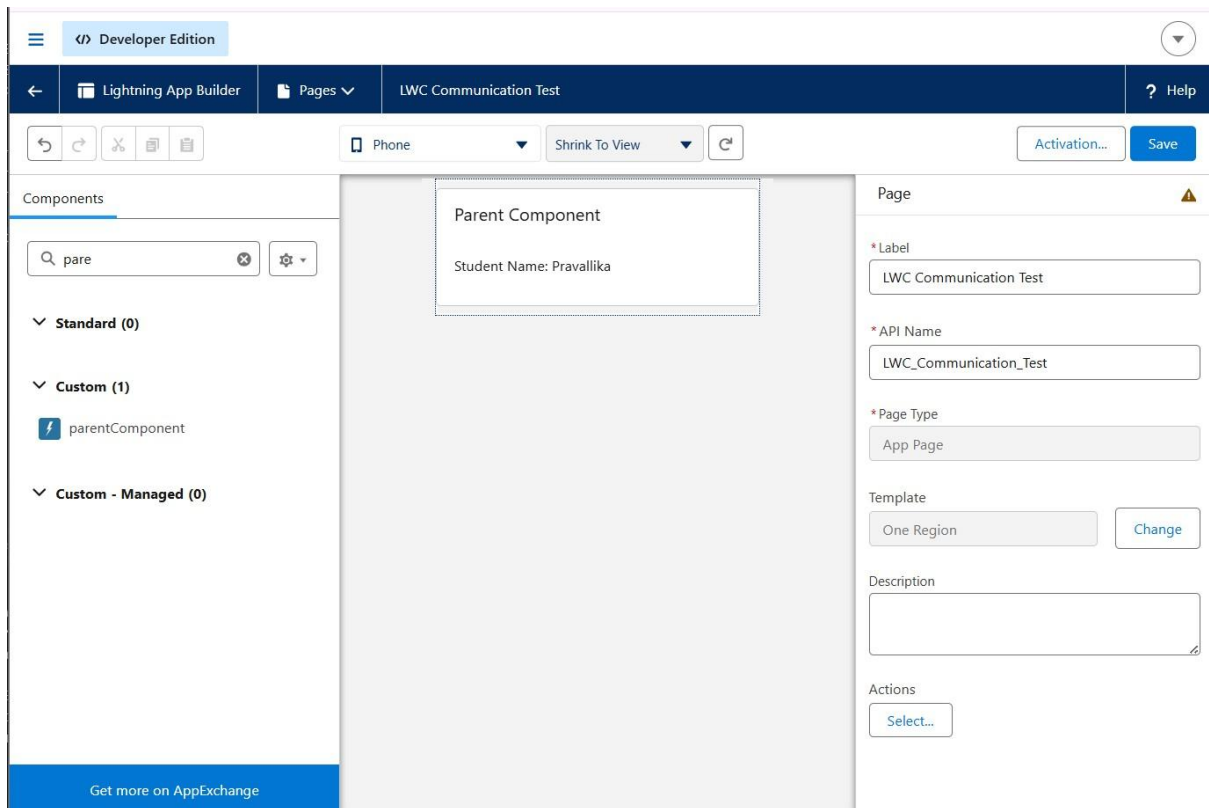
LWC Component Architecture:

A Lightning Web Component application is divided into small, reusable components. Each component handles a specific task.

The architecture used includes:

- Parent Component – Controls the main application.
- Child Component – Handles specific actions and sends information to the parent.
- Job Card – Displays job details and actions.
- Student Profile – Manages student information.

This structure makes the application simple, organized, reusable, and easy to maintain.



3. Parent Component:

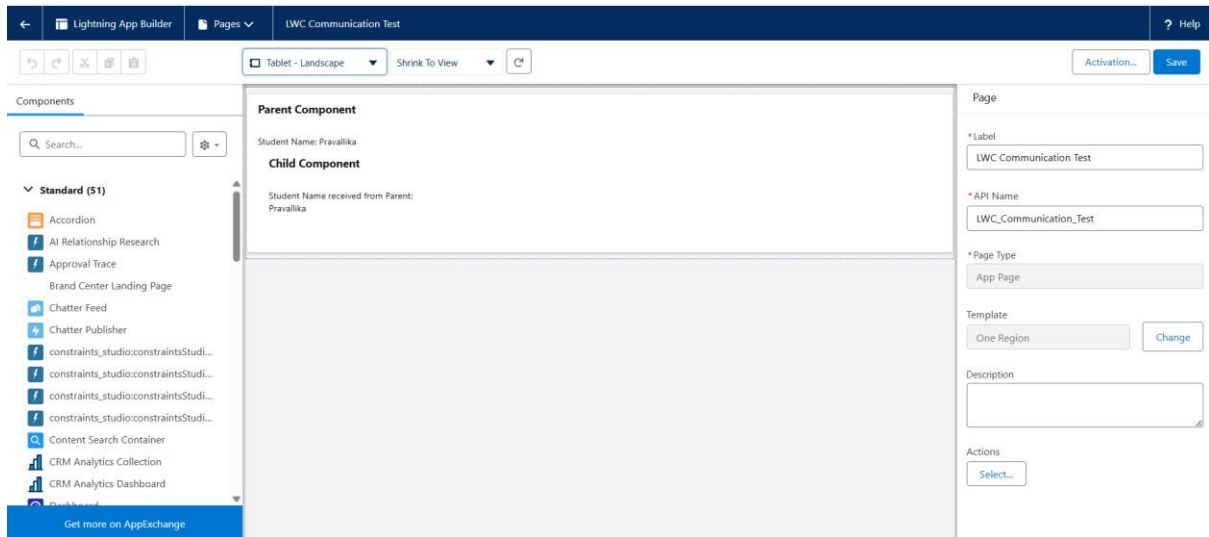
The Parent Component manages the main application and coordinates communication between components. It stores the required student information and handles data received from the Child Component.

The parent is responsible for controlling the overall functionality while the child handles its own specific tasks. This makes the application organized, reusable, and easy to maintain.

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4. Child Component:

The Child Component manages its own interface and user interactions. In today's implementation, it received student information from the Parent Component and displayed it on the screen. A button was also provided in the Child Component to send information or an event back to the Parent Component. This shows how data and events can be exchanged between a parent and child component.

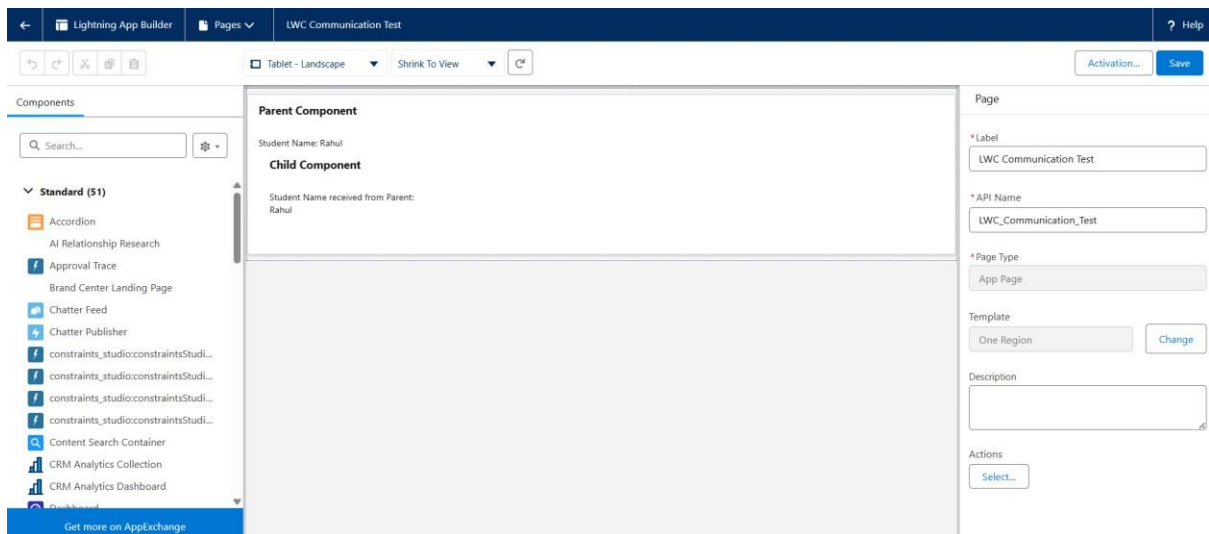


5. Parent-to-Child Communication:

Parent-to-Child communication is used to pass data from a parent component to a child component.

In the application, the Parent Component passes the student name to the Child Component. The child receives the value and displays it on the screen.

This approach provides a simple and clear way to share data between components and keeps the application well organized.



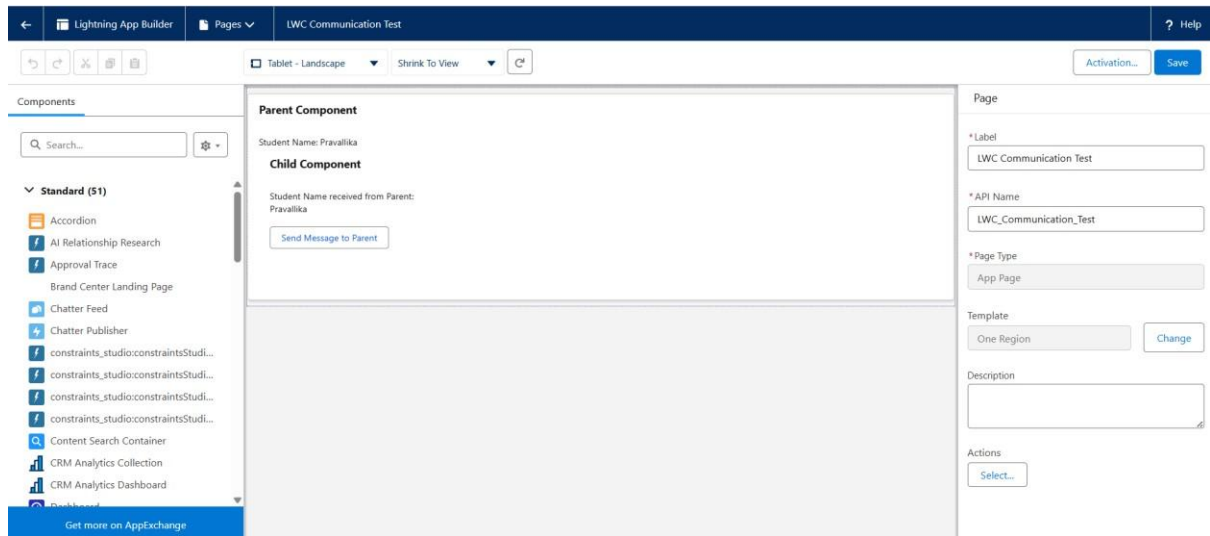
6. Child-to-Parent Communication:

Child-to-Parent communication allows a child component to send information or notify an action to its parent component.

In the application, the Child Component sends an event when the user performs an action. The Parent Component receives the event and uses the information to update the screen.

This helps components communicate clearly and handle user actions efficiently.

Custom Events are commonly used in Lightning Web Components to establish communication from a child component to its parent. This approach provides a simple and organized way for components to exchange information.



7. Custom Events:

Custom Events

Custom Events are used in LWC to send information from a Child Component to its Parent Component.

When a user performs an action, the child creates an event and sends the required information to the parent.

Custom Events can be used for:

- Button clicks
- Job selection
- View Details actions
- Apply actions

The Parent Component receives the event and performs the required action. This keeps component communication simple and organized.

8. Job Card Component

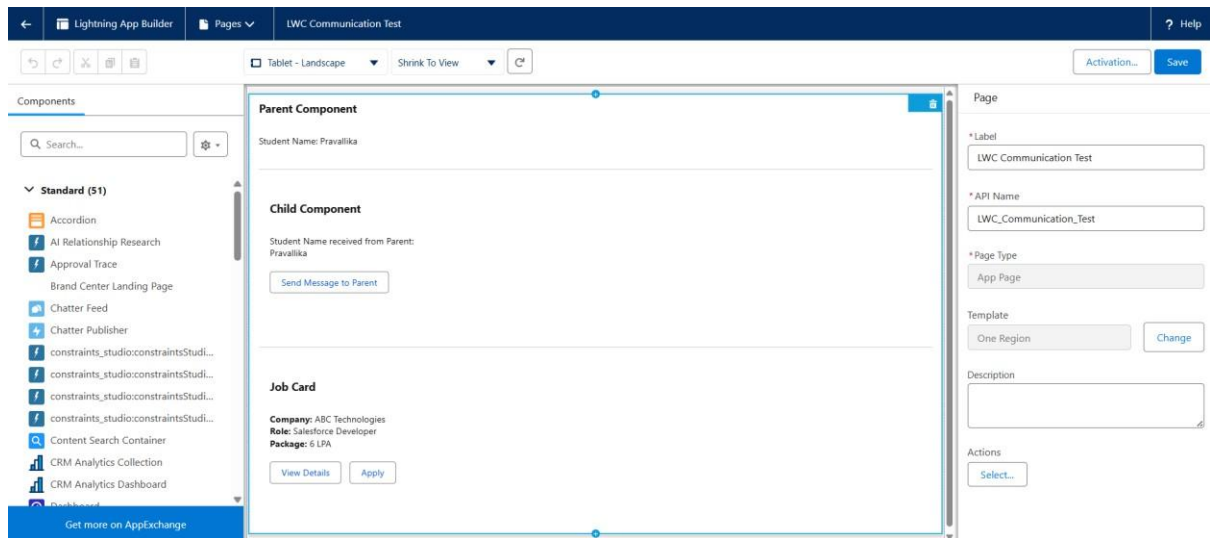
The Job Card Component displays job opportunity details in a simple and organized format.

It includes:

- Company Name
- Job Position
- Salary Package
- View Details

- Apply

The component is reusable, so the same Job Card can display different job opportunities by changing the data. This reduces duplicate code and keeps the application organized.



9. View Details Action:

The View Details button allows the user to view information about a selected job. When clicked, the Job Card sends the job details to the Parent Component using a Custom Event.

The Parent Component receives the information and displays the required details.

10. Apply Action:

The Apply button allows the user to apply for a job. The Child Component sends the apply action to the Parent Component.

The Parent Component handles the next step of the application process. This keeps the components simple and organized.

11. Student Profile:

The Student Profile Component is used to collect and display student information for placement activities.

It includes details such as:

- Student ID
- Phone Number
- Email
- Branch
- CGPA

- Skills
- Preferred Location

This component provides a structured and reusable form for managing student details.

The screenshot shows a web application interface for a 'Student Profile'. At the top, there's a navigation bar with a search bar and several icons. Below the navigation bar, the 'Student Profile' form is displayed. It contains the following fields:

- * Phone:** A text input field containing '9586975243'.
- * Email:** A text input field containing 'padalapravallika@gmail.com'.
- * Branch:** A dropdown menu with 'CSE' selected.
- * CGPA:** A text input field containing '8'.
- Skills:** A text input field containing 'python'.
- Preferred Location:** A dropdown menu with 'Hyderabad' selected.

At the bottom of the form, there is a blue 'Save Profile' button. Below the button, a green message states 'Profile saved successfully!'.

12. Student Profile Fields:

The fields used in the Student Profile are:

12. Student Profile Fields

Field	Purpose
Student ID	Identifies the student
Phone	Stores contact number
Email	Stores email address
Branch	Stores academic branch
CGPA	Stores academic performance
Skills	Stores technical skills
Preferred Location	Stores preferred job location

Required fields: Phone, Email, Branch, and CGPA

Optional fields: Skills and Preferred Location.

13. Lightning Base Components

The Student Profile Component uses Salesforce Lightning Base Components to build its form and user interface. These ready-made components help create a consistent and user-friendly design without developing every UI element from scratch.

The form uses components for:

- Text fields
- Email fields

- Number fields
- Dropdown menus
- Multi-line text areas
- Buttons
- Card-based layouts

Using **Lightning Base Components** helps maintain the standard Salesforce look and feel while providing built-in functionality and consistent behavior across the application.

14. Student Profile Custom Object

A Salesforce custom object named **Student Profile** was created to store the profile information.

The API name of the object is:

Student_Profile__c

The object contains fields corresponding to the Student Profile form.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Branch	Branch__c	Picklist		
CGPA	CGPA__c	Number(3, 2)		
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
Preferred Location	Preferred_Location__c	Picklist		
Skills	Skills__c	Text Area(255)		
Student ID	Student_ID__c	Auto Number		
Student Name	Name	Text(80)		✓

Sure. Here is a copy-paste ready version in the same style:

15. Validation

Validation is used in the Student Profile to make sure the required information is entered before saving.

The mandatory fields are:

- Phone
- Email
- Branch

- CGPA

If any required field is empty, the profile will not be saved. This helps maintain complete and accurate data.

16. Required Field Validation

When the user clicks Save Profile without filling a required field, a validation message is displayed.

The message shown is:

“Complete this field.”

This prevents incomplete student records from being saved.

This confirms that required-field validation is working.

The screenshot shows a web application interface with a top navigation bar and a sidebar. The main content area displays the 'Student Profile' form. The form includes the following fields:

- * Phone**: A text input field with a red 'X' icon and the message 'Complete this field.' below it.
- * Email**: A text input field with a red 'X' icon and the message 'Complete this field.' below it.
- * Branch**: A dropdown menu with 'Select an Option' selected and the message 'Complete this field.' below it.
- * CGPA**: A text input field with a red 'X' icon and the message 'Complete this field.' below it.
- Skills**: A text area.
- Preferred Location**: A dropdown menu with 'Select an Option' selected.

A blue 'Save Profile' button is located at the bottom left of the form.

17. CGPA Validation

CGPA is treated as a numeric field.

The Student Profile restricts CGPA to a valid range of:

0 to 10

This prevents users from entering an invalid academic score.

The validation is performed before the profile is submitted.

The screenshot shows a web application interface. At the top, there is a navigation bar with a blue cloud icon, a search bar, and several icons (star, plus, triangle, question mark, gear, bell with '2', and a user profile). Below the navigation bar, the main header includes 'Playground Starter', 'Welcome', and a dropdown menu with '* LWC Communication Test' selected. The main content area is titled 'Student Profile' and contains several form fields:

- *Phone:** A text input field containing '5484'. Below it, a red error message reads: 'Please enter a valid 10-digit phone number.'
- *Email:** A text input field containing 'njdnkdd'. Below it, a red error message reads: 'Enter a valid email address, such as name@email.com.'
- *Branch:** A dropdown menu with 'IT' selected.
- *CGPA:** A text input field containing '12'. Below it, a red error message reads: 'The number is too high.'
- Skills:** A text input field that is currently empty.
- Preferred Location:** A dropdown menu with 'Select an Option' selected.

18. Save Profile Functionality:

The Save Profile function is used to store student information in Salesforce.

The process is:

Enter Details → Validate Fields → Validate CGPA → Send to Apex → Save Record → Show Success Message

This demonstrates how LWC connects with Apex and Salesforce to save student data.

The screenshot shows a web application interface with a top navigation bar. The navigation bar includes a logo on the left, a search bar in the center, and several icons on the right. Below the navigation bar, there is a breadcrumb trail: "Playground Starter" > "Welcome" > "Install a Package" > "Get Your Login Credentials" > "LWC Communication Test". The main content area is titled "Student Profile". It contains a form with the following fields: "Phone" (text input with value "9586264535"), "Email" (text input with value "padalapravalli@gmail.com"), "Branch" (dropdown menu with value "CSE"), "CGPA" (text input with value "9.50"), "Skills" (text input with value "python, java"), and "Preferred Location" (dropdown menu with value "Hyderabad"). A blue "Save Profile" button is at the bottom of the form. Below the button, a message says "Profile saved successfully!".

19. Apex Integration:

Apex handles the backend operations of the Student Profile.

The two main operations are:

- Save Student Profile – Saves student details in Salesforce.
- Get Student Profile – Retrieves an existing student profile.

The LWC communicates with Apex to perform these operations.

Flow:

Student Profile LWC → Apex Controller → Salesforce Object

20. Load Profile Functionality:

The Load Profile functionality is used to retrieve an existing student's profile from Salesforce.

The student's email address is used to identify the profile.

The retrieved information is then displayed in the Student Profile form.

The process is:

Enter Email

↓

Click Load Profile

↓

Apex searches for the profile

↓

Profile information is retrieved

↓

Information is displayed in the form

↓

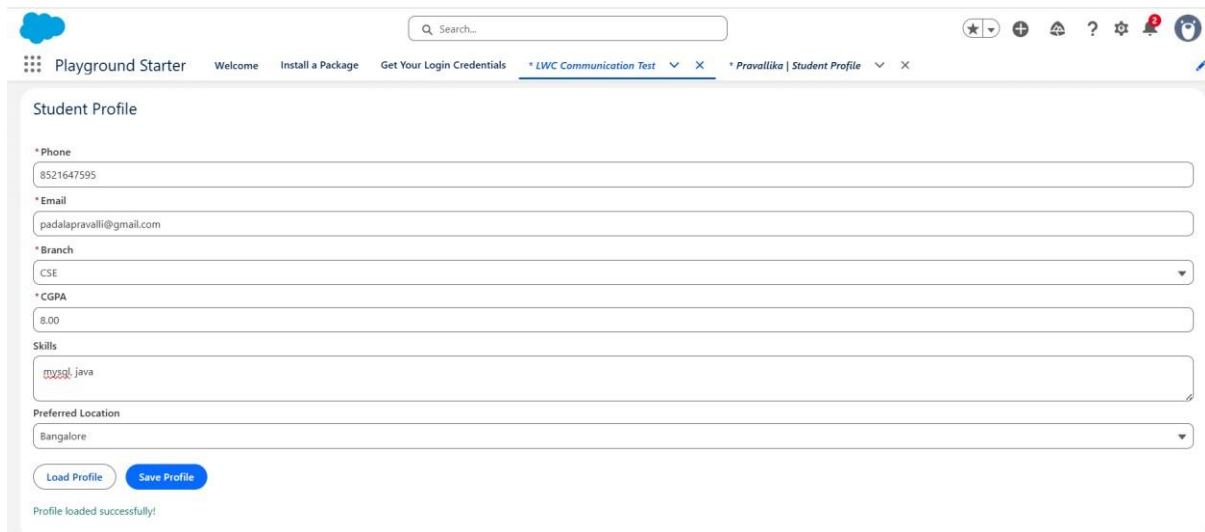
Profile information is returned

↓

Fields are populated

↓

Profile Loaded Successfully



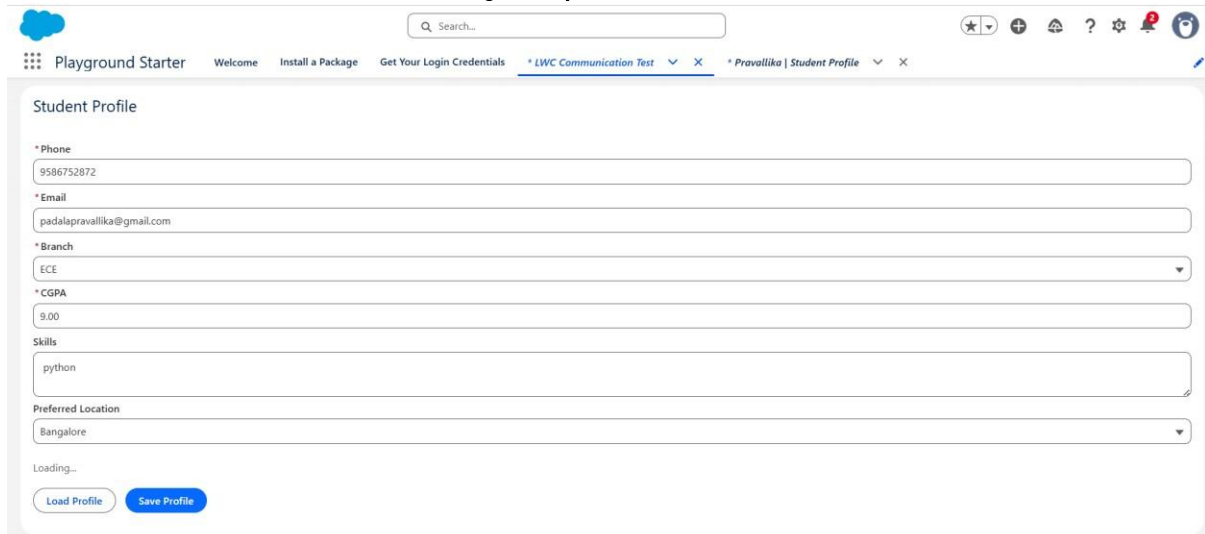
The screenshot shows a web application interface with a top navigation bar containing a search bar and several icons. Below the navigation bar, there are tabs: "Playground Starter", "Welcome", "Install a Package", "Get Your Login Credentials", "LWC Communication Test", and "Pravallika | Student Profile". The "Student Profile" tab is active, displaying a form titled "Student Profile". The form contains the following fields: "Phone" (8521647595), "Email" (padalapravalli@gmail.com), "Branch" (CSE), "CGPA" (8.00), "Skills" (mysql, java), and "Preferred Location" (Bangalore). At the bottom of the form, there are two buttons: "Load Profile" and "Save Profile". A message "Profile loaded successfully!" is displayed below the form.

21. Loading State:

The Loading State is used to show that the application is processing a Save or Load request. It gives the user clear feedback that the operation is in progress and prevents confusion.

The application handles three main states:

- Loading – Request is being processed.
- Success – Operation is completed successfully.
- Error – An issue occurred during the operation.



The screenshot shows the same "Student Profile" form as above, but with a "Loading..." message displayed below the form. The "Load Profile" and "Save Profile" buttons are still visible.

22. Success Message:

After completing an operation successfully, the application displays a confirmation message to the user.

- Save: "Profile saved successfully!"
- Load: "Profile loaded successfully!"

These messages confirm that the requested action was completed successfully and provide clear feedback to the user.

This provides clear feedback to the user.

The screenshot shows a web application interface with a top navigation bar and a main content area. The navigation bar includes a search bar and several icons. The main content area displays a 'Student Profile' form with the following fields:

- * Phone: 9586752872
- * Email: padalapravallika@gmail.com
- * Branch: ECE
- * CGPA: 9.00
- Skills: python
- Preferred Location: Bangalore

At the bottom of the form, there are two buttons: 'Load Profile' and 'Save Profile'. The 'Save Profile' button is highlighted in blue. Below the buttons, a message states 'Profile loaded successfully!'.

23. Error Handling:

Error Handling is used to manage problems during Save and Load operations.

If an error occurs while communicating with Apex, an error message is displayed to the user.

This helps the application remain reliable and user-friendly by clearly informing the user when an operation fails.

Student Profile

*Phone
5484
Please enter a valid 10-digit phone number.

*Email
njdnkdd
Enter a valid email address, such as name@email.com.

*Branch
IT

*CGPA
12
The number is too high.

Skills

Preferred Location
Select an Option

24. Lightning Data Service:

Lightning Data Service (LDS) provides standard Salesforce features for working with records. It reduces the need for custom Apex in common record operations. In this project, Apex was used for the custom Save Profile and Load Profile functions.

25. Reusable Component:

Reusable components can be used in multiple places without creating new components each time. For example, the Job Card can display different job records using different data. This helps reduce duplicate code and improve maintenance.

26. Component Responsibilities:

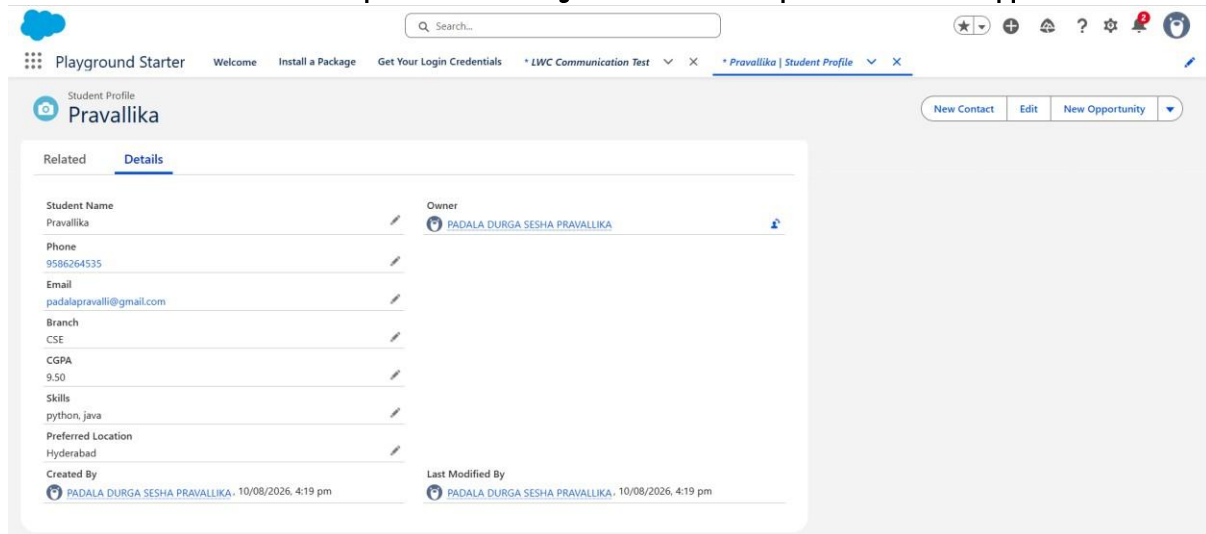
Component	Responsibility
Parent Component	Manages and coordinates components
Child Component	Displays data and sends events
Job Card	Displays job details and actions
Student Profile	Manages student information

Dividing responsibilities makes the application simple and organized.

27. Complete Application Flow:

The overall application flow is:

Student → Student Summary → Student Profile → Save/Update Profile → Eligible Jobs → Job Card
 → View Details/Apply → Applications → Application Status
 This shows how different components work together as one complete Salesforce application.



28. Testing Performed:

The following scenarios were tested during the implementation:

Test	Expected Result
Leave Phone empty	Required-field message
Leave Email empty	Required-field message
Do not select Branch	Required-field message
Leave CGPA empty	Required-field message
Enter invalid CGPA	Validation error
Enter valid profile	Profile saved
Load existing email	Profile displayed
Save successful profile	Success message
Loading operation	Loading indication
Apex failure	Error message

29. Problems Encountered and Resolved:

Issue 1: Save Profile Button Not Working:

The Save Profile button was not saving the student details. The validation, JavaScript event, Apex connection, and Salesforce fields were checked and corrected. After fixing these areas, the profile was saved successfully.

Issue 2: Required Fields Empty:

When required fields were left blank, the message "Complete this field." was displayed. This confirmed that validation was working correctly.

Issue 3: Student Profile Object:

The Student Profile custom object and its fields were checked in Object Manager to ensure they matched the LWC and Apex implementation.

30. Final Result:

The final application successfully demonstrates:

- Parent-to-Child Communication
- Child-to-Parent Communication
- Custom Events
- Job Card Component
- Student Profile Component
- Form Validation
- CGPA Validation
- Save and Load Profile
- Apex Integration
- Loading, Success and Error States
- Reusable Components

Overall, the sprint demonstrates how LWC, Apex, validation, and Salesforce data can work together to build a functional application.